

Week 4 · Laboratory Solutions

- These are **reference** answers, not the only correct ones. The checker tests the physics, not the diagram.
- Read them after attempting the problem.

Building and checking

```
cd lectures/W04_simulink/solutions
W04_S1_heading_loop      % builds W04_S1.slx – all three problems in one model
W04_S_expected           % regenerates the expected-result figures

W04_check(1,'W04_S1')
W04_check(2,'W04_S1')
W04_check(3,'W04_S1')
```

All three pass and `check_overlaps('W04_S1')` is **0**. The output below is the measured one.

```
W04 problem 1
steady error at Kp = 30      0.0362 (expected  0.0000 +- 0.05 deg) PASS
overshoot at Kp = 30        -0.0389 (expected -0.0100 +- 0.25 %) PASS
steady error at Kp = 100     0.0065 (expected  0.0000 +- 0.05 deg) PASS
overshoot at Kp = 100       0.2400 (expected  0.2400 +- 0.25 %) PASS
steady error at Kp = 300     0.0019 (expected  0.0000 +- 0.05 deg) PASS
overshoot at Kp = 300       1.5261 (expected  1.5300 +- 0.25 %) PASS

W04 problem 2
overshoot at Kd = 0  [%]     11.7355 (zeta = 0.3265)
overshoot at Kd = 25  [%]     4.0964 (zeta = 0.5179)
overshoot at Kd = 74.9 [%]   -0.0366 (zeta = 0.9000)
overshoot falls as Kd rises                                     PASS

W04 problem 3
turn WITH the wrap           19.9987 (expected 20.0000 +- 3 deg) PASS
turn WITHOUT the wrap        -339.9969 (expected -340.0000 +- 12 deg) PASS
```

The law, and the three decisions in it

$$\tau_N = K_p ssa(\psi_d - \psi) - K_d r$$

	The choice	Why the alternative is worse
The D term feeds back $\dot{r}$	one Selector on state 6	$d\epsilon/dt$ agrees with $-\dot{r}$ while ψ_d is constant and disagrees at every step , where the command's derivative is an impulse that goes straight into the actuator. Nothing in this model is differentiated
The error is wrapped before the gain	<code>ssa</code> inside the law	Without it a 20° command across the seam is executed as a 340° turn. Problem 3 measures exactly that
Both feedbacks are Goto/From tags	<code>psi_fb</code> , <code>r_fb</code>	Two lines crossing the whole model backwards land on the forward path. <code>check_overlaps</code> is the check

Why the steady error is zero, and why that is not a compliment to the controller

$$M_{66} \ddot{\psi} + (|N_r| + K_d) \dot{\psi} + K_p \psi = K_p \psi_d$$

The heading is the integral of the yaw rate, so the plant carries a **free integrator** and the loop is **type 1**. Week 3's plant had none, and no value of K_p could reach the setpoint there.

The same substitution answers Problem 2 without simulating anything: K_d appears **beside the damping coefficient**, so raising it raises ζ and overshoot falls. In Week 3 the controlled variable was a velocity, its derivative was an acceleration, and the identical term sat beside the **mass**. The term did not change; the axis did.

The layout trap this model walked into

The five constants inside the allocation were laid out at `[90, 60+34i, 150, 84+34i]`. That puts `k_neg`'s centre on $y = 140$ — which is exactly the row the `x_ff` inport feeds along. `check_overlaps` found the two lines drawn on top of each other.

The fix is to offset the ladder so that **no constant's centre lands on an inport row**:

```
'Position', [90 78+34*i 150 102+34*i]
```

The general lesson: when a subsystem has inports at fixed heights and a column of constants beside them, the two ladders have to be interleaved deliberately. Autorouting will not do it, and the diagram looks fine until it is printed.

One measurement subtlety in Problem 3

The checker reads ψ **unwrapped** — straight from state 12, without applying `ssa`. That is deliberate: a wrapped angle cannot tell $+20^\circ$ from -340° , because both end at the same physical heading. The quantity the problem is about is the **turn executed**, not the heading reached.

The two runs also command from $t = 0$ rather than stepping at $t = 5$ s. With a step, the two runs are already in different places when it arrives, and the comparison stops being about the wrap.